EXERCISE 10.5

Choose the correct or the most suitable answer from the given four alternatives.

(1) $\frac{d}{dx}\left(\frac{2}{\pi}\sin x^\circ\right)$ is

(1) $\frac{\pi}{180}\cos x^\circ$ (2) $\frac{1}{90}\cos x^\circ$ (3) $\frac{\pi}{90}\cos x^\circ$ (4) $\frac{2}{\pi}\cos x^\circ$

(2) If $y = f(x^2+2)$ and $f'(3) = 5$, then $\frac{dy}{dx}$ at $x = 1$ is

(1) 5 (2) 25 (3) 15 (4) 10

(3) If $y = \frac{1}{4}u^4$, $u = \frac{2}{3}x^3 + 5$, then $\frac{dy}{dx}$ is

(1) $\frac{1}{27}x^2(2x^3+15)^3$ (2) $\frac{2}{27}x(2x^3+5)^3$

(3) $\frac{2}{27}x^2(2x^3+15)^3$ (4) $-\frac{2}{27}x(2x^3+5)^3$

(4) If $f(x) = x^2 - 3x$, then the points at which $f(x) = f'(x)$ are

(1) both positive integers (2) both negative integers

(3) both irrational (4) one rational and another irrational

(5) If $y = \frac{1}{a-z}$, then $\frac{dz}{dy}$ is

(1) $(a-z)^2$ (2) $-(z-a)^2$ (3) $(z+a)^2$ (4) $-(z+a)^2$

(6) If $y = \cos(\sin x^2)$, then $\frac{dy}{dx}$ at $x = \sqrt{\frac{\pi}{2}}$ is

(1) – 2 (2) 2 (3) $-2\sqrt{\frac{\pi}{2}}$ (4) 0

(7) If $y = mx + c$ and $f(0) = f'(0) = 1$, then $f(2)$ is

(1) 1 (2) 2 (3) 3 (4) – 3

(8) If $f(x) = x\tan^{-1}x$, then $f'(1)$ is

(1) $1+\frac{\pi}{4}$ (2) $\frac{1}{2}+\frac{\pi}{4}$ (3) $\frac{1}{2}-\frac{\pi}{4}$ (4) 2

(9) $\frac{d}{dx}(e^{x+5\log x})$ is

(1) $e^x.x^4(x+5)$ (2) $e^x.x(x+5)$ (3) $e^x+\frac{5}{x}$ (4) $e^x-\frac{5}{x}$

(10) If the derivative of $(ax-5)e^{3x}$ at $x = 0$ is – 13, then the value of a is

(1) 8 (2) – 2 (3) 5 (4) 2

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